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Choosing the right Learning Management System

A requirements-based comparative study at a small consulting firm

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Abstract

There are several hundred Learning Management Systems (LMS) available today. In order to decide which LMS is a good fit for a specific company, it is important to define a list of requirements for an LMS. The purpose of this thesis was to construct a list of requirements for a small consulting firm and test a selection of LMSs. An additional purpose was to evaluate the list of requirements against previous research on the area of digital learning and learning in LMSs. The thesis' result might help other companies to evaluate LMSs in comparison to their needs.

The requirements were found by conducting eleven interviews with employees at the consulting firm and their customers, and performing a thematic analysis on the interview data. The criteria for selecting LMSs was a free trial period and suitability according to the company's business requirements. The LMSs chosen for detailed testing were Teachable, Thinkific, AddWisdom and Learnifier. The result showed that AddWisdom fulfilled all high priority level requirements. However, Teachable and Thinkific fulfilled the majority of the company's functionality requirements. In the analysis, deficiencies in the list of requirements were found, which are important to take into account in the future.

Keywords

Learning Management System, LMS, Comparative study, Digital learning, List of requirements

Sammanfattning

Det finns flera hundra digitala lärplattformar (eng. Learning Management Systems, LMS) tillgängliga idag. För att kunna avgöra vilken digital lärplattform som passar ett specifikt företag bäst är det viktigt att definiera en kravlista för en lärplatform. Syftet med detta examensarbete var att konstruera en kravlista för ett mindre konsultföretag och testa ett urval av lärplattformar. Ett ytterligare syfte var att utvärdera kravlistan mot tidigare forskning på området digitalt lärande och lärande i digitala lärplattformar. Resultatet av uppsatsen kan hjälpa andra företag att utvärdera plattformar i jämförelse med deras behov.

Kraven identifierades genom att genomföra elva intervjuer med anställda på konsultföretaget och deras kundföretag, och genomföra en tematisk analys på intervjudatan. Kriterierna för att välja en digital lärplattform var en gratis provperiod och lämplighet enligt företagets affärskrav. De plattformar som valdes för detaljerad testning var Teachable, Thinkific, AddWisdom och Learnifier. Resultatet visade att AddWisdom uppfyllde alla högprioriterade krav. Dock uppfyllde Teachable och Thinkific majoriteten av företagets krav på funktionalitet. I analysen upptäcktes brister i kravlistan, som är viktiga att ta hänsyn till i framtiden.

Nyckelord

Learning Management System, LMS, Jämförande studie, Digitalt lärande, Kravlista, Lärplattform

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Chapter 1

Introduction

In a world offering an accelerating amount of digital solutions for all sorts of needs, great possibilities exist for transferring education to digital platforms. Today, in the backwater of Covid-19, when a major part of education was forced to be conducted digitally, digital platforms are a key solution in many courses. The motivation, pandemic excluded, behind choosing to include digital tools in education and learning varies between educators and learners. For educators the most popular reason is the good communication possibilities, whereas learners believe that time efficiency is the most important aspect of information technology aided learning (Lonn & Teasley, 2009).

When turning away from the users' opinions and looking into the research field of digital learning, the potential for learning is the main focus. According to Clark and Mayer (2016), learning is defined as an experience that changes the learner's existing knowledge. One aspect that affects distance learning is the so-called *transactional distance* (Moore, 1989). The theory of transactional distance explains how three key factors, *structure*, *dialog* and *autonomy*, will affect learning in any environment by increasing or decreasing the transactional distance. These factors highlight the importance of structuring a course, facilitating dialog between the instructor and the learner and how the autonomy, including motivation and self-activation, of the learner could be favoured (Moore, 1989). Since these are important factors to take into account in distance education in general, they are also important when developing digital distance education.

There are many digital tools to choose from when digitising learning, and one option

is to use a *Learning Management System* (LMS). An LMS is a software system used to structure and distribute digital courses in an interactive environment, with built in tools to allow material distribution to learners, data collection regarding the learners' outcomes and communication between instructors and learners (Turnbull et al., 2020). The use of LMSs is simplifying education in workplaces for companies of all sorts and sizes, allowing an efficient way of educating staff in relevant work related areas (Sabharwal et al., 2018).

There are a multitude of different LMSs available on the market and choosing the one best suited for a specific company can be difficult. According to Sabharwal et al. (2018), important factors to consider are usability, technical characteristics of the LMS, cost-effectiveness, customisability and maintainability. The specific requirements connected to each factor will however always be unique for each company and can thus best be formulated by understanding the needs and desires of the potential setting where the LMS will be used. In this study, the specific needs of a small consulting company who are educating in the field of digital accessibility will be evaluated in order to find a suitable LMS.

1.1 Purpose

The purpose of this thesis is to present which requirements the small consultant firm Axess Lab has on an LMS and to illustrate how different LMSs on the market meet these requirements. The requirements will relate to multiple professional roles, and will describe both functionality and business aspects. In addition to this, the purpose is to discuss how the requirements correlate to previous research in the area of digital learning and learning in LMSs, in order to understand how the requirements can be refined to enhance learning.

1.2 Research questions

In order to satisfy the purpose of the thesis, three research questions have been formulated.

- Which requirements does a small consulting firm and their customers have on a Learning Management System (LMS)?

- In which ways do the requirements on an LMS benefit course participants' learning?
- How does a selection of LMSs meet the requirements in comparison to each other?

1.3 The client

The company, which the study is performed together with, is a small consulting firm called Axess Lab (from now on referred to as *the client*) based in Stockholm, Sweden. At the time of the study, the client has 17 employees (Axess Lab, n.d.). Operating in the field of digital accessibility, the client's three main tasks include (a) building digitally accessible solutions, (b) offering consulting services for companies wanting to develop their knowledge of digital accessibility and (c) educating about accessibility. Various types of education is provided by the client, where the focus is on educating external customers. At the time of the study, the client educates through live lectures and workshops. To make it possible to educate more people with less resources, the client wishes to make their education more efficient through digitalization of the course material (Axess Lab's CEO, personal communication, February 10th, 2023). However, the aim of the client is not to implement only digital courses, some education will still be held live as workshops. The digitisation of the material should be implemented through the use of an LMS, which requires an investigation of which LMS best responds to the client's and their customers' needs and criteria.

1.4 Benefits and sustainability

A large part of the Swedish population turns to digital online services to perform their everyday tasks (Statistics Sweden, 2022). With an increased number of users comes an increased width in the users' abilities, needs and capacity to use these digital tools, making digital accessibility an important issue. The matter of making web pages and apps digitally accessible comes down to a question of inclusion, since this can determine the difference between people being able to handle their everyday life without being reliant on others or not. To ensure accessibility in digital environments there are existing and forthcoming laws, guidelines, and standards (Funka, n.d.). Finding efficient ways to educate and reach more people involved in the process of developing web pages, apps

and other digital services is thus important in the quest for a more equal and inclusive society.

Education within digital accessibility could also be linked to sustainable development. Sustainability is traditionally divided into three dimensions, environmental, social and economic sustainability, where education and accessibility mostly concerns the social dimension. One of the sustainable development goals presented by the United Nations is Goal 9, *Build resilient infrastructure, promote sustainable industrialization and foster innovation* (United Nations Development Programme, 2015c). This goal has a target linked to it which demands increased access to information and communications technology. This could partly be achieved by inclusive digital innovation, where technologies are made available regardless of who the user is. Inclusion in social, economic and political aspects regardless of disability is mentioned in Goal 10, *reduced inequalities* (United Nations Development Programme, 2015a), as well. This could include both the right to be able to participate in social contexts online and access digital information flows, which could be enabled by ensuring the people who are developing these digital environments are aware of how to do it in an inclusive way. To be able to foster innovative thinking and inclusive digital environments, the knowledge required must be shared with concerned parties. In relation to Goal 4, *Quality education* (United Nations Development Programme, 2015b), one way to do this could be done by education through LMSs.

1.5 Delimitations

In this study, a list of requirements has been created based on collected data from eleven interviews with employees at only three companies, the client and two of the client's customers. Because of the time and resource limits of the study a selection of eleven LMSs were chosen for further testing, were four LMSs were thoroughly tested against the list of requirements. In the prototype course created in the LMSs, existing learning material created by the client was used. The user experience testing of the LMSs was performed on a limited group of a few employees of two different customers to the client, because of the limited time resources at these companies.

1.6 Related work

A number of studies have already been conducted with the purpose of comparing different LMSs according to different criteria. In this section highlights from four studies are presented.

Little (2015) writes about practical benefits of LMSs and explores important features and characteristics of LMSs on the corporate market. Little's study contains two case studies describing LMS usage in global organisations as well as an analysis of key issues in the process of buying an LMS for a corporate context. The results point towards the long term benefits of taking the time to investigate which LMS is most suitable for the business in question, and that the choice will affect the organisation as well as the learning.

Karadimas (2018) defines 38 important LMS features and compares ten commonly used LMSs based on these features. The results of Karadimas' study highlight ways in which the LMSs are both different and similar to each other, and that the best suited LMS depends on the user and their specific needs. The LMS which covers most features included in the study is the open source solution Moodle. Another study which points towards the benefits of the LMS Moodle is a study performed by (Sk et al., 2021). In their study, they chose six open source LMSs to analyse, where Moodle proved to be the one meeting most criteria. Subramanian et al. (2014) performs an examination of the two LMSs Moodle and Blackboard. The results are based on data from three case studies, and the focus areas for the comparison are communication, productivity and student involvement tools. In the comparison Moodle scores better than Blackboard.

As presented, some studies have been conducted on the area of finding an LMS for a workplace. However, these are the only up-to-date studies found by the authors, which points at a possible gap in the research area. Furthermore, Little's research (2015) indicates long term benefits of performing a thorough investigation before choosing an LMS. This investigation might require non-existent economical resources for a small company, and therefore this study will not only help the client in their decision, but it might also be a good resource for other small companies in their search for a suitable LMS.

1.7 Outline

This thesis consists of four additional chapters, Theoretical background, Method, Results and Analysis and discussion. In the theoretical background a detailed overview of research in the field of digital learning and learning in LMSs is presented, which is later used in the analysis. In Method the methodology of the study is presented together with detailed descriptions of the different methods used and motivations for these choices. The methods used include semi-structured interviews, thematic analysis and a comparative study of LMSs. Presented in Results is the list of requirements which was established from the themes created in the thematic analysis. Following this, the highlights of the comparative study are illustrated. In Analysis and discussion the results are analysed and discussed with the purpose of answering the research questions, which culminates in a conclusion and directives for future research.

Chapter 2

Theoretical background

Learning management systems are developed as products utilising information technology for education and learning, in addition to providing tools for administration and organisation (Turnbull et al., 2020). Since one of the main functions of an LMS is to provide tools which can ease the learning process, a deepend study in the area of digital learning and learning in LMSs is justified. In this chapter, important concepts and factors related to learning are presented.

2.1 Important design aspects in an LMS

When educating in an LMS it is vital to ensure that the LMS is user-friendly, in order to avoid the scenario where the learners spend more time learning how to use the system than understanding the material (Sabharwal et al., 2018). User-friendliness in an LMS is highlighted by Sims (2021) who declares that it is critical to test the LMS beforehand, otherwise learners might encounter errors when undertaking an online course, which might affect their motivation and attitude towards learning. A study performed on several universities shows that when educators and developers take time to develop user-friendly software for digital learning it enhances the learning experience for the learners, which positively affects the learners' results (Bailey et al., 2018). Another possibility to enhance the learners' experience in an LMS is to ensure that the system language of the LMS is the same as the language used in the content of the courses (Little, 2015). In addition to this, it is advantageous to ensure that an LMS can be accessed on mobile platforms, which lowers the threshold to start courses (Sabharwal et al., 2018).

Furthermore, in an LMS, it is vital that multiple sorts of content elements can be included. Examples include videos, podcasts, written lectures, quizzes, internet articles, blog posts and presentations (Sims, 2021). The purpose of the usage of multiple types of content is both to engage the learner and to encourage critical and analytical thinking. Veluvali and Surisetti (2022), presents multiple studies indicating that when audio-visual support, such as video content, is included in curriculum and course content, learning increases. Another way to improve learning is to encourage the learners to repeat the content in a meaningful way (J. R. Anderson, 2010). A possible way to enable repetition opportunities for the learners is to allow the learner to revisit the material even after course completion.

2.2 Course design

Learners have more difficulty keeping motivation and discipline when taking online courses compared with face-to-face courses (Bailey et al., 2018). To assist and help learners through an online course it is important to set a clear and detailed curriculum with specific learning objectives. Similarly, a clear learning path has a positive impact on learners' learning (Sims, 2021). When creating a course in an LMS it is often easy to establish a learning path with a curriculum and learning objectives, where it is obvious for the learners what the next step in the course is and what the outcome should be. However, even in a course with a clear learning path, problems might arise. If the material that the learner meets is too complex, it creates another possible pitfall in digital learning, where the material might exceed the cognitive capacity of the learners (Clark & Mayer, 2016). A potential solution for avoiding this scenario is to divide the material into smaller modules, which could create an easier learning path for the learners. Furthermore, the use of weekly deadlines in an online course can have a positive impact on the learning process, since the deadlines can help the learners to evenly distribute their work (Fulton et al., 2013).

One risk with digital learning is that the cognitive process of the learner's generative processing might be insufficient, if the learner lacks motivation to engage fully with the material (Clark & Mayer, 2016). Using varied content forms such as visuals and learning activities could be one solution to engage the learner. Important to keep in mind when creating different types of content formats is however to keep the intended learner and their needs in focus, since the learners are the most important users of the

LMS (Sabharwal et al., 2018; Sims, 2021). Also, Sims (2021) implies that accessibility needs and learners' preferences must be taken into consideration when choosing content forms for online courses. Another possible thing to include is *digital badges*. The use of digital badges in online courses can improve the learners' motivation, when the learners receives badges for their accomplishments (Iwata et al., 2019). That being said, there are other studies indicating that even though the learners might feel more motivated when receiving badges, it does not change the performance level of the learners (Balci et al., 2022).

Another interesting possibility with digital learning is the concept of *gamification*. Gamification is defined by Deterding et al. (2011) as the use of elements in a course design which shares similarities with game functionality but are in a non-game context. These elements can for example be direct feedback, positive reinforcements and level-graded tasks, which has the purpose of not only enabling interaction between the learner and the material, but also giving the learner the feel of playing a game (Veluvali & Surisetti, 2022). Gamification is promising to use when developing online courses (Veluvali & Surisetti, 2022), and studies have shown a positive correlation between gamification and learners' academic results (Hamari et al., 2014). This is positive, since studies show that learners require stronger motivation when attempting digital learning (Rawashdeh et al., 2021; Veluvali & Surisetti, 2022).

2.3 Communication and evaluation

In order to improve learning and determine if the education enables learning, it is important to evaluate the learning progress (Veluvali & Surisetti, 2022). To perform this evaluation one has to gather information from different learners' progress, and mining data from an LMS is an effective way to accomplish this. According to Bailey et al. (2018) it is critical to evaluate the impact of the education on learners' progress and results in order to achieve successful online courses, which is only possible if the instructors gather information about the learners' progress. In digital learning tools real-time data can be extracted, which provides an effective way to develop courses and secure high-quality education (Bailey et al., 2018). Tools like quizzes, discussion forums and questionnaires are often used in LMSs and the data collected in these are useful when analysing the academic process of the learner group (Veluvali & Surisetti, 2022). In an LMS it is possible to assess the skills of a learner in several ways, for ex-

ample by utilising the possibility of including multiple kinds of quizzes (Govindasamy, 2001). The commonly used multiple-choice question is often easy to implement in an LMS and can be an effective way to assess a learner's learning progress even in high-level education.

Feedback is a powerful tool to excel learners' learning, where direct and personal feedback and instructions related to learning objectives has been discovered as the most efficient form of feedback (Hattie & Timperley, 2007). Hattie and Timperley (2007) has pinpointed three questions that are important to acknowledge when giving feedback, (a) "*Where am I going?*", (b) "*How am I going?*" and (c) "*Where to next?*". When developing online education it is critical to ensure quality feedback to the learners, especially since there is no in-classroom face-to-face communication during the course (Leibold & Schwarz, 2015). When developing the feedback procedure in digital courses for colleges and universities, Bailey et al. (2018) claims that one important practice is to provide *differential support* in order to ensure high quality and learner success in fully online learning. One possible action to provide the differential support is to individualise the feedback given to the learner from the instructor (Bailey et al., 2018). Another simple tool to confirm to the learners that they are on the right path is to include digital badges which are received when completing parts of a course (Iwata et al., 2019). On top of that, an important aspect is to ensure that the learner's progress through the course is visualised. This will help the learner to understand and be aware of how they are doing in a specific course (Klerkx et al., 2014).

Except for sharing feedback, an LMS also makes it easier for the instructor to communicate and share different types of information with the learners (Sims, 2021). The enhanced communication in digital learning (compared with traditional education forms), was the most common answer amongst instructors to the question of what the greatest advantage of using information technology in courses is (Lonn & Teasley, 2009). When it comes to interaction, Moore (1989) defines three different kinds of interaction important in digital learning, *learner-instructor interaction*, *learner-content interaction* and *learner-learner interaction*. Learner-instructor interaction is defined as where the instructors,

> First having planned or been given a curriculum, a program of content to be taught, they seek to stimulate or at least maintain the student's interest in what is to be taught, to motivate the student to learn, to enhance and

> maintain the learner's interest, including self-direction and self-motivation. (Moore, 1989, p. 3)

A popular tool to enhance the learner-instructor interaction is for the learner to receive regular reminders from the instructor (Martin & Bolliger, 2018). Learner-content interaction is defined as when the learners "'talk to themselves' about the information and ideas they encounter in a text, television program, lecture, or elsewhere." (Moore, 1989, p. 2). Without this interaction there can be no education, and when learners intellectually interact with the content it changes and develops the learners' understanding. This type of interaction where learners are getting self-taught is the most common strategy for adult learners (Moore, 1989). Lastly, learner-learner interaction is when learners interact with each other, in example peer reviews and analytic group discussions. T. Anderson (2003) summarises the relation between these interaction forms with the concept of *equivalence theory*, which he defines as

> Deep and meaningful formal learning is supported as long as one of the three forms of interaction (student–teacher; student–student; student–content) is at a high level. The other two may be offered at minimal levels, or even eliminated, without degrading the educational experience. High levels of more than one of these three modes will likely provide a more satisfying educational experience, though these experiences may not be as cost or time effective as less interactive learning sequences. (T. Anderson, 2003, p. 4)

Thus, each kind of interaction can play a vital role in the digital learning experience. However, not all three interaction types need to be present at the same time, even though it might enhance the learning experience, since it risks affecting the time or cost effectiveness of the course.

Chapter 3

Method

In this chapter the general methodology and research process for the study is presented. It is followed by a detailed description of the methods used to answer the study's research questions, together with comments of the credibility of the methods in question.

3.1 Methodology

The study was carried out using a *qualitative* methodology. The main point of a qualitative study is that the research methods for collecting and analysing data are designed for data in the form of continuous text, with the purpose of achieving a holistic view (Denscombe, 2016). This differs from a *quantitative* study, where the data is mostly composed of numbers and a few specific variables are analysed (Denscombe, 2016). Furthermore the advantages of data collection in a qualitative study is that the data extracted is colourful and nuanced (Denscombe, 2016). These advantages align well with the purpose of this study, see Section 1.1, which is why this methodology has been chosen.

In a qualitative study the researcher might be seen as a tool, whose purpose is to process the collected data (Denscombe, 2016). In the context of this study the authors have been utilised as tools when performing a *thematic analysis* and when testing the LMSs. The use of qualitative research methods can also give the authors the possibility to process and refine the research questions during ongoing data collection (Denscombe, 2016; Maxwell, 2008). This possibility of refining the research questions fits well for

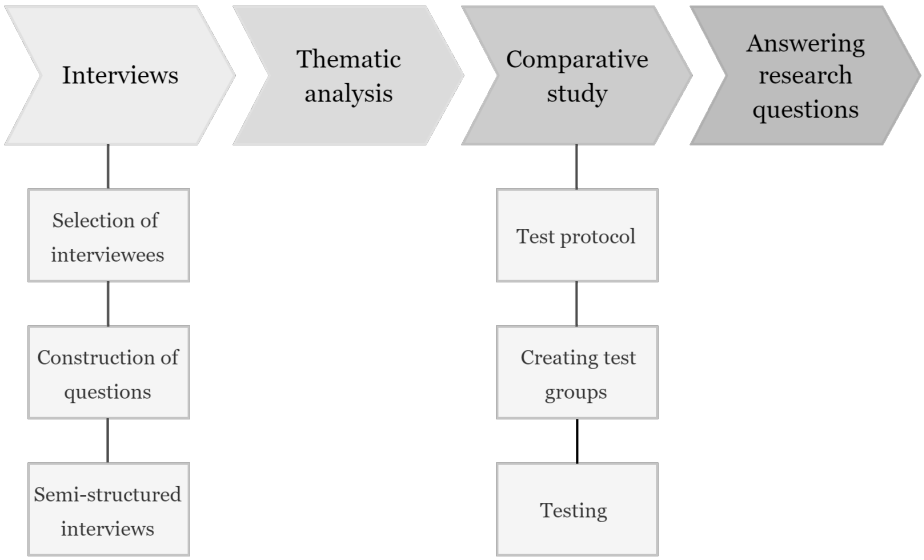
this thesis since the relationship with the data and the client developed over time.

The disadvantages of using a qualitative study method are that the *generalisability* of the result from the data collection and analysis might be low (Denscombe, 2016). The reason for this is the often limited number of individuals that are deeply studied, which is why according to Maxwell (2008) one can speak of *transferability* instead of generalisability in qualitative research. In the light of this study it is not possible to generalise the results to all companies, but the result might be transferable to match with companies of similar size with similar interests.

3.2 Research process

Figure 1

Illustration of the research process



To answer the research questions, a series of different methods have been used, which are illustrated in Figure 1. Empirical data was collected by performing semi-structured interviews, which required construction of interview questions and selection of interviewees. The interview data was then analysed with an implementation of thematic analysis described by Braun and Clarke (2006). The analysis produced a list of themes which was translated into a list of requirements for an LMS. The list of requirements was then used in a *comparative study*, where a selection of LMSs were tested following a test protocol derived from the requirements. An evaluation of the list of requirements and the test results was then performed with the purpose of answering the research

questions.

3.3 Interviews

During the study a total of ten semi-structured interviews were performed, using the Zoom video conferencing tool. Present at each interview were one interviewer, one notetaker and one or two interviewees. The interviews were conducted in Swedish and recorded using the built-in function of Zoom.

3.3.1 Semi-structured interviews

Semi-structured interviews are common in qualitative studies and are interviews where the interviewer has pre-constructed a list of questions that are asked during the interview (Denscombe, 2016). Apart from these questions the interviewer is also able to ask the interviewee follow-up questions with the purpose of gaining a deeper understanding of the subject at hand. This differs from a structured interview where the questions asked during an interview are strictly bound to the list of questions constructed beforehand. Furthermore, when performing a semi-structured interview series, the authors can add questions to the list between the interviews if for example new information is introduced during one interview. This allows the authors to explore new areas of the study during the interview series (Denscombe, 2016). A semi-structured interview is common to use when the authors want a deeper understanding of a subject and where the interviewees might not have full awareness of the subject at hand (Kallio et al., 2016).

These prerequisites conforms well with the purpose of this study, where the interviews have the purpose of giving the authors a deeper understanding of the interviewees' requirements on an LMS, which might be a subject that the interviewees have limited experience of. The possibility to change and develop the questions during and between interviews is also an advantage, due to the width of possible areas of expectations that the interviewees might have. A possible pitfall when using interviews is that the validity of the data extracted from the interviews might be low (Denscombe, 2016). The reason for this is that people do not always do what they say they do, which can affect the results. To avoid this pitfall and to keep the validity of the data high, it is good practice to validate the data during the interview by asking follow-up questions. During the interviews in this study, follow-up questions were asked multiple times, for example

by rephrasing an opinion the interviewee expressed to make sure that the interviewer understood it correctly. To avoid misinterpretation of the words of the interviewees it is also important to implement some version of *respondent validation*, where the authors systematically gather feedback on the data set from the people studied (Maxwell, 2008). To validate the results of the interviews, the list of requirements was sent to the client to be reviewed and confirmed.

3.3.2 Selection of interviewees

To make the first selection of interviewees an *explorative selection* method was used. An explorative selection is where researchers try to find edge cases in the population studied, and is commonly used in qualitative research, where it fits particularly well for small research studies (Denscombe, 2016). This differs from a *representative selection* where the researchers try to imitate a whole population in the research group and the interviewees are a cross section of the group studied. The former selection method was selected for this study, because of the purpose of evaluating what the specific client's needs and requirements are, which can be difficult to generalise to a larger population.

To achieve an explorative selection of interviewees, the sub-method of *subjective selection* was used. Subjective selection works with the principle that the best information will be had from handpicked interviewees (Denscombe, 2016). The criteria for being handpicked is that the person is relevant for the researched subject and that they have some specific knowledge in the area that might be relevant. This method differs from a systematic selection, where for example every third person on the list of employees is chosen for an interview.

When deciding the selection of interviewees in qualitative research one must always keep in mind the importance of *data saturation*. Saturation in this context means that the selection must give the possibility of reaching a point in the data collection where there are no new themes or codes coming forward (Fusch & Ness, 2015). To achieve data saturation the *cumulative approach* was used to define the number of interviewees. A cumulative approach is where the authors expand the selection of interviewees if data saturation has not been achieved (Denscombe, 2016).

For the purpose of this study, six relevant roles to interview were distinguished. The chosen roles can be seen as edge cases because of their different expertise on the subject

of digital education in LMSs and their roles in relation to the client. The six roles chosen to be part of the study were:

- CEO
- Educator
- Developer
- User-experience designer (UX-designer)
- Marketer
- Course participant

The client was consulted during the subjective selection for these different roles, to enable a selection where the interviewees could give the authors different perspectives of the subject. To broaden the data collection and to achieve data saturation it was decided to interview at least two people for every role. This was achieved for every role except for the role CEO, see Table 1. Since the client is a small company, it is common that one employee covers more than one role and therefore some interviewees were interviewed for several roles. After the scheduled interviews two more interviews were added, since it was not possible to detect if data saturation had been achieved. During the last interviews no new themes emerged, which is a sign of data saturation according to Fusch and Ness (2015).

Table 1

Interviews and roles of the interviewees

Interview number	Role of interviewee
1	UX-designer and marketer
2	Educator and marketer
3	Educator and UX-designer
4	Educator
5	Educator and UX-designer
6	CEO
7	Educator

Continues →

Interview number	Role of interviewee
8	Course participant
9	Two course participants
10	Course participant

3.3.3 Construction of questions

The questions used during the interviews were constructed following the guidelines and recommendations in *Questions and answers – question design in self-administered- and interview questionnaires* (Statistics Sweden, 2016). Initially, a single interview guide was designed, where the questions were divided into segments according to which role the question was intended for. One section consisted of general questions, relevant to every role. The questions in the interview guide were all centred around three main themes: the client's current education, the interviewee's experience with any previously used LMSs and expectations on a prospective LMS. The questions were participant-oriented and formulated using words like *what*, *how* and *who*, which according to Kallio et al. (2016) increases the chances of receiving descriptive answers, which was the goal. When creating the interview guide for each individual interview, appropriate segments from the interview guide were included based on which role(s) the interviewee had. The interview guide consisting of all segments can be found in Appendix A.

Before the interviews were conducted, the interview guide and its questions were discussed with supervisors, both from the client and from the university. During these discussions some formulations were changed and questions were both added and removed from the interview guide. Besides the predetermined interview questions of the interview guide, spontaneous follow-up questions were used during the interviews. As mentioned by Kallio et al. (2016), follow-up questions can be used in order to direct the interviewee towards areas more closely related to the research questions. In addition to spontaneous follow-up questions added during interviews, some questions were added to the interview guide after interviewees mentioned something which was deemed to be worth investigating further. One example of this was when one interviewee mentioned the possibility of receiving a certificate after a completed course in an LMS, so in the coming interviews after this, all interviewees were asked about their opinion on

this functionality.

3.4 Thematic analysis

To analyse the data set from the semi-structured interviews, a *thematic analysis* was performed with the purpose of constructing a list of requirements for an LMS. In this section an overview of the theoretic process of a thematic analysis will be presented, together with descriptions of how the thematic analysis was performed in context of this study.

Thematic analysis is a widely used data analysis method, which has the purpose of finding and analysing themes in a set of data which relates to the research questions (Braun & Clarke, 2006). According to Kiger and Varpio (2020) thematic analysis is appropriate to use when searching for shared meaning and thoughts between people, which aligns well with this study's research questions. Another similar analysis method is *qualitative content analysis*, which also has the purpose of defining themes in a data set (Mayring, 2000). Qualitative content analysis could also have been used to answer the study's research questions, but there is an important difference between the two methods which affected the choice for this study. Even though the process of the two methods are similar, one major difference is that in qualitative content analysis it is important to take in consideration the frequency of different themes and code when reviewing the themes (Vaismoradi et al., 2013). In the context of this limited study it is difficult to rely on the quantity of data points, which makes it more reasonable to choose a more qualitative approach with the help of thematic analysis.

According to Braun and Clarke (2006) a thematic analysis can be used as an *essentialist thematic analysis* where the analysis reports the reality of the interviewees and their experiences and meanings. This differs from the *constructionist thematic analysis* which has the purpose of theorising the socio-cultural contexts behind the answers from the interviewees. In this study an essentialist approach was chosen, in view of the fact that the thematic analysis will be used to produce a list of requirements for an LMS. Braun and Clarke (2006) has structured the method into six phases, which are presented down below together with details about the specific phase and important notes about critical parts of the phase.

Phase 1: familiarising yourself with your data

During the first phase the data set was read through multiple times. Notes were taken on parts of the data set that could be interesting for the next phases. The data set should consist of transcribed interviews according to Braun and Clarke (2006). However, the authors of this study did not transcribe the interviews. The reason for this is twofold: firstly, both authors were present during all interviews, which meant that one author could lead the interview and fully focus on the interviewee while the other author was able to take detailed notes. Second, since the purpose of the interviews (to collect requirements for testing the LMSs) was clear throughout all interviews, the risk of missing vital information which could only be detected through transcription was deemed as very small.

Phase 2: generating initial codes

In the next phase the data was coded to include the most basic segments or elements of the data set, which in a meaningful way related to the area studied. Braun and Clarke (2006) explains that coding can be done for either semantic or latent themes. A semantic analysis approach is where the themes created from the codes are only explaining the surface of the data set, what the interviewee actually has said, and not the meaning behind it (Braun & Clarke, 2006). In contrast, if the authors decide to find latent themes in the data set, they try to go beyond and examine the subcontext behind the words of the interviewee. The purpose of the thematic analysis for this study was to create a list of requirements, which corresponds to the creation of semantic themes. Thus, in this second phase the semantic content of the data set was coded.

When coding, the authors looked for parts of the data set that would be helpful when constructing the list of requirements, but also coded other parts of the data which did not directly correspond with the research questions. This approach can be analysed in relation to *inductive* versus *theoretical* analysis. According to Braun and Clarke (2006) an inductive analysis is performed when the codes do not map against a pre-existing coding frame. When performing a theoretical analysis the codes will instead correspond directly to the authors' theoretical interest on the subject and give a more detailed view on a specific part of the data set (Braun & Clarke, 2006). The coding in this study has thus been a combination of both inductive and theoretical analysis.

Phase 3: searching for themes

During the third phase of the thematic analysis all the codes from phase 2 were divided into themes. A theme is according to Braun and Clarke (2006) an important part of

the data that relates to the research questions. This shows a crucial difference between a qualitative and a quantitative study, where if a quantitative analysing method had been used it would be important to count the numbers of codes related to one theme (Braun & Clarke, 2006). Each theme was divided into multiple sub-themes, where each sub-theme related to the main theme it belonged to. .

Phase 4: reviewing themes

During this phase all themes were reviewed to ensure data coherence within the themes. In addition, all themes were compared against each other in order to verify clear enough distinctions between the themes. One purpose of phase 4 is to determine if the thematic map gives an accurate reflection of the data set (Braun & Clarke, 2006). This is why the original data set was reviewed once more to ensure no important data points had been missed in the earlier phases.

Phase 5: defining and naming themes

As described by Braun and Clarke (2006), this is the phase where the themes and sub-themes are defined and refined to determine the essence of every theme. This was carried out by going back to the data set which had been extracted for the theme and organising it with the purpose of telling a background story for the theme. Additionally, the themes were analysed to determine how they correlate to the research questions. The sub-themes were refined by creating requirements corresponding to each sub-theme. Each requirement was given a priority level, decided by the client.

Phase 6: producing the report

During the last phase of the thematic analysis the result from previous phases was taken and formatted to be a part of this report's results, with the purpose of giving the reader a full understanding of the themes and their relations to the data set.

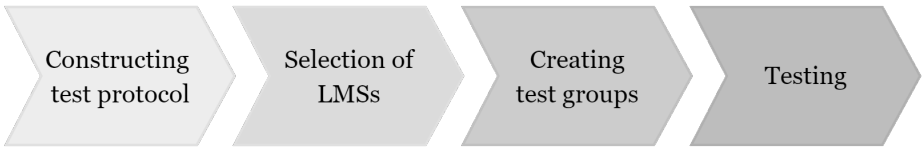
3.5 Comparative study

One of the main goals of the study was to analyse chosen LMSs with regards to the client's needs. In order to do so, a test protocol was constructed from the requirements found through the interviews and a selection of LMSs was done. User test groups were created, consisting of potential course participants. Each LMS was then tested according to the constructed test protocol. The tests were performed by the authors, the client

and user test groups. The whole process of the comparative study can be seen in Figure 2.

Figure 2

Illustration of the comparative analysis process



3.5.1 Constructing test protocol

For each requirement on the requirement specification, a test case was designed. Since the main purpose was to check whether a requirement was fulfilled or not, the test cases were designed in a way which allowed simple *yes/no* answers. However, a requirement is most often not completely fulfilled, meaning the functionality might not be implemented in the best way imaginable. Considering this, all test cases had an evaluation scale with numbers between 0–3, where 1 equals a functionality that exists in the LMS but is barely usable, 2 equals a functionality which has some flaws or disturbing limitations, and 3 equals a functionality which is implemented in a satisfying way. 0 equals that the functionality could not be found in the tested LMS. To test the requirements concerning usability, the *System Usability Scale* (SUS) developed by Brooke (1995) was used. SUS is a globally used *Likert scale* and has a high face validity in measuring the subjective usability of a software system. The scale is constructed using ten questions, see Appendix B, that measure the experience users have when using a specific system. The score of a single respondent is calculated with this specific formula:

> To calculate the SUS score, first sum the score contributions from each item. Each item's score contribution will range from 0 to 4. For items 1,3,5,7,and 9 the score contribution is the scale position minus 1. For items 2,4,6,8 and 10, the contribution is 5 minus the scale position. Multiply the sum of the scores by 2.5 to obtain the overall value of SU. (Brooke, 1995, p. 6)

The total score of the system, which has a range from 0 to 100, is the average value of all the respondents' submitted SUS scores.

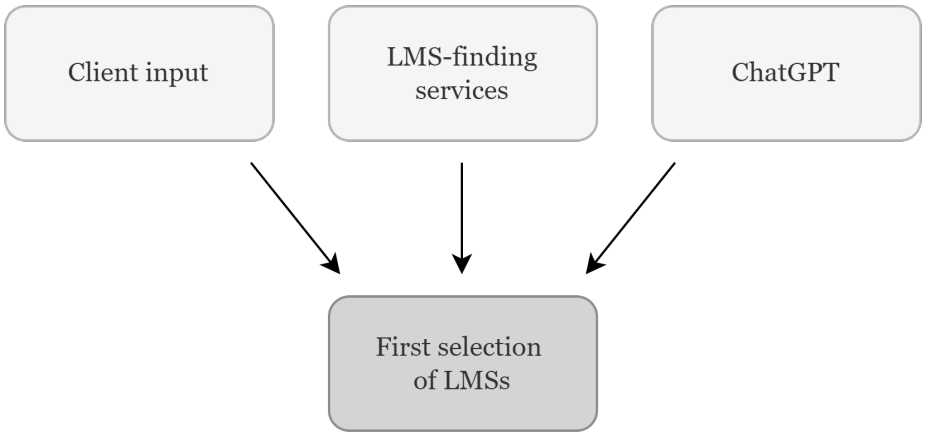
A test protocol was designed by combining the test cases with simple instructions. This was done in order to ensure equivalent testing of each LMS. The protocol was divided

into segments corresponding to the different user roles (educator, course participant or the client). Implementing the introductory course in digital accessibility was included in the test protocol, as part of the course creator experience. The test protocol can be found in Appendix C.

3.5.2 Selection of LMSs

Figure 3

Initial LMS selection process



There exist hundreds of different LMSs on the market (Hennigan & Main, 2023) and it is not possible to test them all. In order to find a wide selection of different LMSs to test, the method of *triangulation* was used in the selection process. According to Patton (1999), a study relying on a single method has a higher risk of producing biassed results. In this study, one risk was that only heavily marketed LMSs would be discovered in the initial investigation of available LMSs. In order to avoid this, three different sub-methods were used to create an initial list of LMSs available. The first submethod was to let the client suggest LMSs that they were interested in. The second submethod of the selection process was to use digital marketplace platforms for evaluation and selection of software. These types of platforms claim to provide services helping businesses and individuals finding the best software, including LMSs. It is however not always clear which LMSs are included in their process and why, which is why two different platforms were used, Capterra Inc. (n.d.) and BusinessWith (n.d.). Both platforms explicitly said that some of the LMSs on their websites paid for advertisement which gave these LMSs a more viewable position. Nevertheless, the platforms also claimed that LMSs that do not pay for advertisement were also shown on the platform. The third and last sub-method was to let the AI-powered chatbot ChatGPT (OpenAI,

n.d.) suggest suitable LMSs to include in the study. This was done by asking chatGPT the question "Could you give me 10 learning management systems for companies available in Sweden and with a free trial period?" and then including the LMSs provided by the chatbot. By combining the suggestions from these three sub-methods, different perspectives were included, which according to Patton (1999) is one way to achieve triangulation.

From the first list of LMSs, all alternatives which did not offer a free trial period longer than seven days were excluded. In addition to excluding alternatives due to monetary limitations, alternatives were excluded according to the business requirements with the highest priority. The filtering with regards to the business requirements was done in collaboration with the client's CEO.

3.5.3 Creating test groups

In order to test the LMSs, one test group for each LMS was created. The test groups consisted of existing customers to the client, who all intended to take the introductory course in digital accessibility. After agreeing to take the course while also testing the usability of an LMS, each participant was asked to fill out a Google form. The form consisted of the following five questions:

- What is your profession?
- What are your main tasks at work?
- What is your education?
- How long have you been working with IT? Reply "0" if you're not working with IT.
- Do you have any other experience/knowledge that you think would affect your experience when you are using and evaluating an LMS?

The reason for collecting information about the participants beforehand was to create equivalent test groups. In the form no sensitive personal information was collected, except for the participants' email addresses. Collecting the email addresses was a necessity in order to be able to send out the right sign-up link to the right test participant. The data containing the email addresses was only accessible by the authors and deleted as soon as the study was concluded. Since each test group would only test one LMS, it was

important that the groups were as similar as possible, so that the result for each LMS could be compared. When creating the test groups, the concept of *matched-groups design* was applied on the criteria *profession/work tasks*, *education* and *experience*. According to the American Psychological Association (n.d.), *matched-pairs design* is a method in which pairs of participants are created, where one participant belongs to the experimental group (exposed to a manipulation) and the other participant belongs to a control group. Matched-groups design works by the same idea as matched-pair design, but allows more than two groups.

Another reason to focus on creating equal test groups is *internal validity*. According to Patino and Ferreira (2018), internal validity concerns the likelihood that the result is a result of what is studied (and not some other cause). In this case, it would mean that if one LMS scores better in the tests, it would be because of how the LMS is designed and not because of the test participants' differences in experience and knowledge. Another version of validity is *external validity*, which Patino and Ferreira (2018) describes as how likely it is that the result of the study is applicable outside the studied group. Since the test groups of this study consisted of real customers to the client, it is likely that future course participants share the view of the usability of the LMSs in this study.

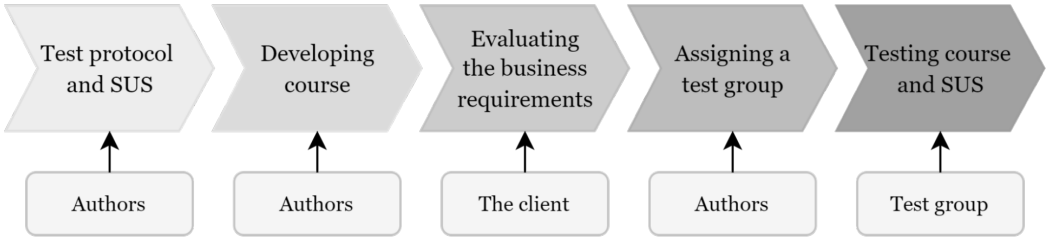
3.5.4 Testing

The testing of each LMS was carried out following a specific procedure, see Figure 4. First, the instructions of the test protocol were carried out by the authors. This included filling out the test protocol with numerical evaluations and descriptive comments, as well as filling out a SUS form from the perspective of the course creator and creating the course in the LMS. Employees at the client helped filling out the evaluation of the business requirements. The second part of the testing consisted of assigning a test group to the LMS. Each participant got instructions for the testing, together with links to the course implemented in the LMS and to a Google form. The form contained questions from the SUS. The participant instructions and the SUS form can be found in Appendix D and Appendix B.

In order to evaluate the business requirements, the client was consulted. Knowledge about the company and their specific needs was judged as vital in order to determine the LMSs' capability to meet requirements concerning customer safety, sellability, drift and sustainability from an economic point of view. The responsibility of testing the

Figure 4

Illustration of testing process of an LMS



requirement concerning digital accessibility was also given to the CEO, and the testing was delegated to an in-house UX-designer. Since the client is an expert in the field of digital accessibility, their assessment was deemed as very reliable.

3.6 Answering the research questions

The first research question of this study, *Which requirements does a small consulting firm and their customers have on a Learning Management System (LMS)?*, was answered by interviewing employees at the client and employees of the client's customers. The interview data was analysed by using a thematic analysis, resulting in themes which were translated into a list of requirements. The second question of the study, *In which ways do the requirements on an LMS benefit course participants' learning?*, was evaluated by using previous research and literature from the field of digital learning and comparing it to the test results. The third and last research question, *How does a selection of LMS's meet the requirements in comparison to each other?*, was answered by conducting tests in a comparative study, where the LMSs were tested against the list of requirements and then later on compared to each other.

3.7 Ethical considerations

Since the data of the study was collected through interviews and forms some ethical considerations were taken into account. According to the Swedish research council (2002), there are four main criteria to obey when conducting research with human participants. The criteria concern: *information*, *consent*, *confidentiality*, and *utilisation*. The *information criteria* states that all participants should be informed of the purpose of the study, the purpose and terms of their participation and any information

about the study that might affect their willingness to participate (Swedish research council, 2002). In this study,this was done by giving each interviewee a short oral presentation at the beginning of the interview. A similar presentation was included in the distributed forms. The second criteria, the *consent criteria*, means that participant participation is voluntary throughout the whole study. Participant consent was collected through an oral question at the beginning of each interview and a written question at the end of each form, together with information given about the participant's right to cancel their participation at any point. The third requirement, the *confidentiality requirement*, declares that all information collected should be stored in a safe way and the fourth requirement, the *utilisation requirement*, states that the information should only be used for research purposes (Swedish research council, 2002). Since no sensitive information was collected in this study, no extra precautions were taken except for storing the data in a cloud environment only accessible by the authors. Also, no personal data about the participants were stored except for email addresses for test participants and first names for interviewees. All data was anonymised in the report and all information was deleted after the study's completion. The data collected was only used for this study.

Chapter 4

Results

In this chapter, the results are presented in two sections. The first section contains the list of requirements created from the thematic analysis of the interview data set. The second section presents the selection of LMSs that were chosen for further testing and a summary of the test results for these LMSs.

4.1 Results from thematic analysis

This section contains the results originating from the thematic analysis of the interview data. Four themes are presented along with sub-themes and requirements linked to each theme. The requirements are categorised as high, medium or low priority and together they form a requirement specification which the LMSs are evaluated against. Along with the sub-themes and requirements, relevant quotes from the interviews are presented, translated from Swedish to English. In Table 2, the respondent number together with the role of the respondent is presented, which allows the reader to trace the presented quotes. The quotes will either highlight the requirement as a whole, or parts of the requirement. Support for each requirement is however always found in the interview notes, even if not presented in the form of a quote in the tables below. Each theme is based on user roles, which in the interviews were mentioned as important in the context of using an LMS. Together the themes build the story of needs and wishes from business, course creator, course participant and customer perspectives.

Table 2

Respondent numbers and roles of the respondents

Respondent	Role of respondent
R1	UX-designer and marketer
R2	Educator and marketer
R3	Educator and UX-designer
R4	Educator
R5	Educator and UX-designer
R6	CEO
R7	Educator
R8	Course participant
R9 & R10	Course participants
R11	Course participant

4.1.1 Theme 1: business requirements

The first theme is based on interview data displaying needs and requirements from a business perspective of the client. Several codes relating to this theme can be traced back to the interview with the client's CEO and many of the requirements are classified as high priority. The sub-themes within this theme are not connected to aspects of usability or the pedagogical possibilities of the LMS, but rather aspects related to business profitability and customer security.

Table 3

Business requirements with high priority

Sub-themes	Requirements	Quotes
Sustainability from an economic point of view	The client's costs of using the platform is a fraction of the revenue	"The most important role I have may be to ensure that this [using an LMS for selling courses] will be economically sustainable, we must be able to sell courses in a way which makes customers buy the courses. Not sustainable to create an LMS only for on-boarding. It is important to be able to build a business model around." - R6
Easy maintenance	Any employee of the client can handle the platform maintenance	"The maintenance usually resolves itself, but the main thing is that several can do it. We usually talk about the 'hit-by-the-bus-scenario'. If I disappear tomorrow, can somebody else take over the maintenance of the website then?" - R3
Safe management of customer data	The servers handling customer data are placed within the EU (GDPR requirement)	"We expect secure management of personal data." - R8
Individual user accounts	Individual user accounts can be created	"Require that there are some kind of specific user accounts." - R6

Continues →

Sub-themes	Requirements	Quotes
Integrated sellability	A payment solution is integrated in the LMS, allowing customers to pay for courses without involvement from the client	"Preferably it should not be required to have an add-on for payments, but preferably it should be integrated in the LMS. So they [the customers] sign up for payment and then it comes in every month." - R6
	Customers can sign up independently without help from the client	"I'm thinking that you go on to our [the client's] website, create an account and do your thing." - R1
	Customers can complete the course independently without help from the client	" [...] all payment and account management is an important part in working with an LMS, that it is taken care of there. Payment, account management, people should be able to add their credit card. A person should be able to register their colleagues and add people there [in the LMS]." - R2

Table 4

Business requirements with medium priority

Sub-themes	Requirements	Quotes
Integrated marketing	Courses are searchable for customers outside the client's customer base	"A wish is also that the LMS comes with integrated marketing." - R6

Continues →

Sub-themes	Requirements	Quotes
Digitally accessible	Able to navigate with the help of keyboard	"Good accessibility level. Embarrassing otherwise. Even if we can not take full responsibility it would be nice if it works." - R5
	Good colour contrast	
	Possible to use screen narrator	

Table 5

Business requirements with low priority

Sub-themes	Requirements	Quotes
Adaptable design	The design is adaptable to the client's theme used elsewhere	"Good if you could change the colours, logo and so on, so it suits our [the client's] 'style'." - R3
Administrative overview	The client is able to see all customers and trace course purchases	"We would definitely like to see who has bought our content/courses. That is a must." - R6
Integrable with other LMSs	The LMS is integrable with other commonly used LMSs	"Then it is interesting if courses could be integrated into our [the customer's] existing LMS." - R8

4.1.2 Theme 2: course creator requirements

During the interviews with the client's educators and UX-designers different aspects of creating courses were brought forward, from the perspective of a course creator. One important focus was on efficiency and easiness when creating and updating course material in the LMS. Furthermore, different requirements for functionality in an LMS were stated, which would give the course creators creative freedom when creating courses.

Table 6

Course creator requirements with high priority

Sub-themes	Requirements	Quotes
Modules and sections	Courses can be divided into modules	"Must be pretty short and concise. Rather a lot of small bits than 'long and thick'." - R4
	Modules can be divided into sections	

Table 7

Course creator requirements with medium priority

Sub-themes	Requirements	Quotes
Easy to updatecourses	Created modules can be used in multiple courses and be updated simultaneously	"Let us say that it is a course in semantic HTML, then it might be one small module about area, one about landmarks and so on. Then you should be able to cut'n'paste, so one can take that module and use it in another course." - R7
	A common material bank exists, where updates on the material is shown in all courses using the material	
Easy to create courses	Visual interface rather than text- or code-based	"Should be easy to assemble courses." - R8
Support for multiple material formats	Images	"It [the LMS] must contain images, must contain videos and not just text." - R1

Continues →

Sub-themes	Requirements	Quotes
	Videos	
	Text	
	Presentations	
	Quizzes	“There is often a need for many people to verify that they have done it right. I think it is a goal you would like to have - the possibility to test oneself.” - R4
User-friendly	Scores high on System Usability Scale	“High points on the System Usability Scale and few/no severe issues when the LMS is user tested.” - R2
	Few/no severe issues	

Table 8

Course creator requirements with low priority

Sub-themes	Requirements	Quotes
Tailored user experience	Notifications settings can be changed	“An important thing in an LMS; possibility to turn of or control notifications.” - R7
	Dark mode can be turned on	“First of all, that you can use dark mode. I always want to use it if it is available.” - R7
Downloadable material	An entire course or parts of it can be downloaded	“Or if it becomes a requirement that we [the client] can hand out the raw material or a manuscript to customers to put in their own LMS.” - R6

Continues →

Sub-themes	Requirements	Quotes
Certificates/badges	Certificates and/or badges can be added to a course	"A badge in the system or a PDF that one can share on LinkedIn. People usually want to put that in their resume." - R6
	Requirements for acquiring a certificate and/or badges are changeable	
Quiz settings	Quizzes can be made mandatory	"If it was possible for me to choose to move on [from the module] I would not have done the quiz. So it is good that it [the quiz] is mandatory." - R7
Deadlines	Deadlines can be added to a course	"Reminder 'Now you can do this' and everything one has to do is to press the link and do what it says." - R6
Reminders	Auto-generated reminders can be sent out to course participants	

4.1.3 Theme 3: course participant requirements

The third theme is built from expectations, or esteemed expectations, from course participants. This means that the quotes supporting the sub-themes and requirements within this theme can be either from real course participants expressing expectations, or from educators at the client, expressing things believed to be important to course participants in their current customer base. A large part of the interviewees' expectations were centered around the possibility to easily track one's process and be able to move freely inside the course with ease.

Table 9

Course participant requirements with medium priority

Sub-themes	Requirements	Quotes
User-friendly	Scores high on System Usability Scale	"High points on the System Usability Scale and few/no severe issues when the LMS is user tested." - R2
	Few/no severe issues	
Visualised process	The course participant's process in a course is visualised	"I would definitely like to see how large the body of knowledge is and how far I have come." - R6 "One always has this progress bar for how many percent one has gone through. I have appreciated it incredibly much because maybe you are tired and then if you see that you have already done 80 percent you get a little cheered on." - R7
Flexible course participation	The course participant's progress is saved	"Possibility to save one's work, go back and take stuff again if wanted." - R1
	The course participant can go back to earlier modules	
	The course participant can repeat completed courses	"Possibility to repeat it [the course], take it again in a pretty easy way." - R8

Continues →

Sub-themes	Requirements	Quotes
Overview of courses	An overview of all courses available to the course participant exists	"[...] easy to see which courses are relevant for me." - R8
Responsive design	The LMS is compatible with mobile platforms	"Work on all platforms (for example to sit with your phone on the subway)." - R1

Table 10

Course participant requirements with low priority

Sub-themes	Requirements	Quotes
Video speed	The playback speed of videos can be changed	"Want to watch at a high speed, the course leader might talk a bit slow." - R2
Tailored user experience	Notification settings can be changed	"An important thing in an LMS; possibility to turn of and control notifications." - R7
	Dark mode can be chosen	"First of all, that one can use dark mode. I always want to use it if it is available." - R7
Discussion forum	Questions can be asked and discussions can be started in a forum	"Then maybe a reminder will come regularly, that you can ask your questions here. And maybe also that one can see others' questions and answers, because you learn from reading what others have asked and gotten as answers." - R10

Continues →

Sub-themes	Requirements	Quotes
Certificates/badges	A certificate and/or a badge can be generated for a course	"So either an undemanding quiz or that when you have gone through all the parts you might receive a certificate. Good if the system could generate this, so you could show it to colleagues or so." - R5

4.1.4 Theme 4: customer admin requirements

The last theme created from the data set was the role of admin at the client's customers. This role is especially important for creating efficiency for the client, where the customers can control their own administration of course participants. Additionally, in the interviews with employees in the position of manager it was clear that it would be helpful to track the course process of course participants.

Table 11

Customer admin requirements with medium priority

Sub-themes	Requirements	Quotes
Overview of course participants	A detailed overview of course participants and their progress in different courses exists	"What is good in my other role [manager] is that in the LMS one can follow up with users who have completed a course." - R8
Administration	New employees can be added and given access to different courses	"Surely there is also a need that some accessibility lead in the company who cares for the LMS, can control how many that have access, pay for the courses and so on." - R2

4.2 Test results

In this section the first selection of LMSs is presented, followed by a motivation of the second selection of four LMSs chosen for further testing. Then the results of the testing of the LMSs are presented.

4.2.1 Selection of LMSs

In the initial selection process, a list of 54 LMSs was produced. After removing all LMSs with shorter free trial periods than seven days, a total of eleven LMSs remained, see Table 12. After an initial evaluation of the highest prioritised business requirements, four LMSs remained for further testing. Moodle was sifted out because it is an open-source LMS, which did not meet the requirement of easy maintenance. Blackboard and Brightspace are LMSs suited for universities and schools and because of this the cost of the platform would not be economically sustainable for the client. Litmos required a separate plug-in to manage integrated payment, which combined with the use of cloud storage in the United States did not meet the business requirements. iSpring Learn and Synap were only suitable for onboarding training and did not have any integrated payment solutions. uQalio was decided not to be sustainable from an economic point of view because of the fee of five dollars for every course participant. The four LMSs remaining for more detailed testing are marked with an asterisk (*) in Table 12.

Table 12

Selection of LMSs

LMS	Can be found at
* AddWisdom	https://www.addwisdom.se/
Blackboard Learn	https://www.blackboard.com/
Brightspace	https://www.d2l.com/en-eu/
iSpring Learn	https://www.ispringsolutions.com/ispring-learn
* Learnifier	https://learnifier.com/
Litmos	https://www.litmos.com/solutions
Moodle	https://moodle.org/

Continues →

LMS	Can be found at
Synap	https://synap.ac/
* Teachable	https://teachable.com/
* Thinkific	https://thinkific.com/
uQualio	https://uqalio.com/

Selection of LMSs sorted in alphabetic order, * marks the remaining LMSs

4.2.2 Testing against the list of requirements

In this section, key points from the results of the testing of the four LMSs Learnifier, Thinkific, Teachable and AddWisdom are presented. The presentation follows the same partition as in Section 4.1, divided according to which user role the requirements relates to. The LMSs received similar grading scores on many of the requirements but there are some requirements where the functionality in the LMSs had major differences. These differences are pointed out and explained, with the purpose of giving the reader a better understanding of the background of the scores and how the LMSs have been graded. The detailed test result for each requirement can be found in Appendix E.

4.2.2.1 Business requirements

The results of the business requirements testing show that most of the LMSs performed quite well and equal on the high priority requirements. When looking at the high priority requirements, Learnifier stands out with a low score on three of the requirements. Teachable and Thinkific have a low score on only one high priority requirement and AddWisdom scores high on all high priority requirements. Learnifier is built to be used as an onboarding tool, but can be changed to use for selling courses. However, the pricing model is not fit for selling to a larger group and therefore Learnifier is very expensive when using it as a platform for selling courses. Both Learnifier and AddWisdom have cloud platforms in Sweden which fulfils the GDPR-requirement, whereas Teachable and Thinkific use cloud storage in the United States. Since Learnifier is not designed to sell courses it does not have an integrated payment solution, but a payment plug-in can be used to make it possible to sell directly in the LMS. None of the LMSs had built in marketing in the platform. This means that a course participant of a differ-

ent course (created by another company) in the LMS does not have the possibility to browse the LMS and find the client's courses. When testing the three medium priority requirements related to digital accessibility, Thinkific scored highest of the four LMSs. Teachable scored slightly higher than Learnifier in the digital accessibility aspect, and AddWisdom scored the lowest. *Possible to use screen narrator* was the accessibility requirement which had the lowest average score when looking at all four LMSs. The LMSs received similar scores on the low priority requirements, however it was not possible in any of the LMSs to test if the LMS was integrable in another LMS.

4.2.2.2 Course creator requirements

The only high priority requirements for the course creator perspective were whether a course could be divided into modules, and if the modules could be divided into sections. All LMSs fulfilled these two requirements. However, none of the LMSs had the functionality of simultaneous update, which was of medium priority, where one module could be used in several courses and if one of these modules were changed the others would be automatically updated. Furthermore, Learnifier had the feature *material bank* where for example an image could be uploaded and used in several modules in one course, and edits on the image in the material bank could be seen in the places the image was used. The other LMSs did not have this functionality.

There were some differences found when testing the LMSs against the requirements with medium priority. When it comes to different kinds of materials, all LMSs supported images, videos, texts, quizzes and presentations. In Teachable, Thinkific and AddWisdom, the presentations were scrollable directly in the lesson, whereas in Learnifier it was only possible to download the presentation. The scores from the SUS performed by the thesis authors showed that Teachable was the most user-friendly with a score of 96, AddWisdom had the second highest score of 88 and Thinkific had a score of 83. Learnifier scored lowest, with a score of 76. The issues found when testing the LMSs from the course creator point of view were that when editing courses in Learnifier, new tabs opened for different editing features, which slowed down the editing process. The editing process in Thinkific was also inefficient because of slow refreshing when switching between different sites in the editing mode. AddWisdom was easy to use, however there were several page titles that were not self-explanatory, which affected the usability. Furthermore, the documentation for AddWisdom was insufficient and customer service had to be contacted to find a specific functionality. No severe

issues were found when testing Teachable.

The scores for the low priority requirements were similar for all LMSs, but there were some differences. Notifications settings were only available in Learnifier, the other LMSs used notifications but did not allow the user to change what to be notified about. None of the LMSs had the option of editing in dark mode and it was not possible to download an entire or parts of course in any of the LMSs. All LMSs had the possibility to add certificates and/or badges to a course, but it was not possible to change the requirements to achieve a specific certificate. In all LMSs the receivement of certificates and badges was connected directly to the requirements of completing a course. Furthermore, all the LMSs had the possibility of making quizzes mandatory, which could be used as a way to regulate the requirements to obtain a certificate or a badge. In Teachable one could make none or all of the quizzes mandatory and in Learnifier the course participants could still skip a mandatory quiz, if they chose not to open the quiz-tab that was linked to in the lesson. In Learnifier both reminders and deadlines could be added. In AddWisdom the expiration date for a course could be changed, which could be interpreted as a way of setting deadlines. Furthermore, it was also possible to send out auto-generated reminders weekly to course participants in AddWisdom. Teachable and Thinkific did not have the functionality of reminders nor deadlines.

4.2.2.3 Course participant requirements

In this category, no requirements were defined as high priority. In the usability testing Thinkific scored highest with a score of 92. Learnifier and AddWisdom followed with the score of 83 and 80, respectively. Teachable scored lowest with a score of 73. Two interesting opinions expressed by the participants, which concerns all LMSs, was that in the LMSs the system language was different than the language used in the course and that when using the LMS one had to create an account specific for the LMS instead of using Single Sign-On (SSO). In Teachable two severe issues were found by the participants. Firstly, the answers in the quiz-section were hard to read. Secondly, the notes taken with the notation functionality could not be found after exiting the notation section. In Thinkific, one issue was that videos started automatically when entering a lesson, and the course participant could not change this setting. When logging in and out of AddWisdom the progress was not saved for one participant and there was also a confusing “Complete lesson”-button in the lessons, which the course participant had to press for the progress to be updated. This button was accompanied with a “Next

lesson”-button which allowed the user to go to the next lesson, but then the progress was not saved.

When excluding the usability aspect, the LMSs met the course participant requirements in a similar way. One functionality in this category that had a large variation between the LMSs was the requirement concerning the course participants’ possibilities of asking questions and interacting with each other. Discussion forums existed in all LMSs but the design choices of the forums resulted in different grades, where AddWisdom had the functionality from the course creator point of view but it was impossible to find when logged in as a course participant.

4.2.2.4 Customer admin requirements

In the different LMSs a user could be assigned to different roles with different permissions. This resulted in a varied score for the requirements for a customer admin. Thinkific did not have a user role that could both see an overview of course participants and add new employees, the latter functionality was restricted to user roles with more course responsibility. AddWisdom had a user role *Group administrator* which fulfilled both requirements perfectly and in Teachable it was possible to customise a user role which fulfilled both requirements. In Learnifier an overview of the course participants’ progress was available for an admin user role, however, the overview was deemed as not detailed enough.

Chapter 5

Analysis and discussion

In this chapter the results of the study are analysed and discussed. First, different aspects of the list of requirements are analysed, including the transferability, strengths, weaknesses and how well the list aligns with previous research on digital learning and learning in LMSs. Then the results of the comparative study are analysed in order to illustrate how the LMSs compare to each other in relation to the requirements. Further, the methods chosen for the study are discussed and aspects that might have affected the results are presented. Finally, a conclusion is drawn and directions for future research are suggested.

5.1 The list of requirements

The list of requirements, formed from the result of thematic analysis, contain aspects related to learning in an LMS, but also requirements which are purely functional or business related. In this section the transferability together with strengths and weaknesses of the requirements are discussed, as well as how aligned the requirements are with previous research in the field of digital learning and learning in LMSs.

5.1.1 Transferability of the requirements

As mentioned in Chapter 3, the results of qualitative research is spoken of as transferable rather than generalisable. In this section, the transferability of requirements is discussed, in relation to the study's research question *Which requirements and expectations does a small consulting firm and their customers have on a Learning Manage-*

ment System (LMS)?.

One example of transferability is the business requirement concerning GDPR, which is likely to be relevant for other companies, since GDPR compliance applies to all businesses. The maintenance aspect is another requirement which is likely to be relevant for other companies of similar size as the client, as small companies rarely have the funding to be able to spare an employee working full time with LMS maintenance. When it comes to evaluating the cost versus revenue from using an LMS, the requirement is not as easily transferable to other businesses, as it depends on how many users the business is able to reach with its LMS. If the cost of the LMS is always proportionate to the number of users, the cost versus revenue evaluation would be transferable. However, since some LMSs have a fixed cost, the economic sustainability requirement would have to be evaluated in relation to each individual business' budget. Thus, there is some uncertainty regarding how this requirement can be transferred.

Requirements concerning sellability, integrated marketing and traceability of customer purchases are mostly transferable, since a goal for companies often is to increase revenue. The possibility to add the company's design to the LMS is also an important factor for many businesses, in order to strengthen the company brand and thereby increase revenue. The requirement of integrability is transferable to other companies with big customers, who have their own LMS in place. Since course creators or instructors of a course do not always have coding experience the requirement concerning a visual interface for the course creator is also transferable. Using a visual interface can be related to usability, which is an important transferable aspect for all user roles. The three requirements linked to digital accessibility should be general for most companies, at least in theory, since there are laws and guidelines concerning accessibility that companies are obliged to follow (Funka, n.d.). Even though the accessibility requirements are transferable, other companies might not have rated these requirements as medium priority.

As presented, many of the requirements can be transferred to other companies of similar size and interest as the client. However, when studying other companies, the prioritisation level of each requirement might differ. A highly prioritised requirement in this study might be just medium priority to other companies, and vice versa. Also, some requirements of the client might not be requirements of another company.

5.1.2 Evaluation of requirements in the context of testing

When testing the LMSs, some of the requirements were found to be flawed, incomplete, and/or unclear. In this section these deficiencies and alternative ways to define the requirements are discussed.

Some requirements received the highest possible score in all LMSs and almost appeared as an established basis for the LMS structure in general, since all tested LMSs had implemented it. One such requirement was the ability to divide material into modules, and divide modules into smaller sections, which was distinguished as important needs of the users and thus formulated into high prioritised requirements. In relation to the need to divide the material into smaller fractions, something else might have been testable in order to differentiate the module functionality in the LMSs from each other. One example could be to test in which way the modules and sections are visualised to the learner. To differentiate the LMSs from each other regarding the module and section functionality is of interest, since these are the only requirements from the course creator perspective rated as high priority by the client. Another requirement which seemed as an established basis was the requirement concerning individual user accounts. When performing the usability tests, one user did however point out that the user should not be required to create an account in order to use the LMS, since users do not want to create unnecessarily many accounts in multiple services. To exclude the individual user account requirement altogether would however result in an impossibility to fulfil some of the requirements concerning the course participant role, such as being able to save user progress. One interesting aspect in order to satisfy both the user not wanting excessive amounts of user accounts and also satisfying the course participant requirements would be to test the possibilities of implementing SSO-login in the LMS. Another solution to the issue of having to create multiple user accounts for different services could be if the LMS was integrable into the customers' own LMS, since no external login would be required. However, the ability to integrate the LMS in an existing LMS was not testable within the limits of this study and could thus not be evaluated as a possible solution.

A medium prioritised need is that the LMS should be able to handle images, videos, text, presentations and quizzes. The requirement for each material format is however limited to only examining whether the material format was insertable in the LMS or not. Here, the requirements could have been expanded to include expectations on the

functionality of the material form, for example whether an image could be edited in the LMS or if subtitles could be shown for videos. Since these functional aspects of the material forms were not distinguished in the thematic analysis, they were not included in the list of requirements. The fact that no further testing on the functionality of each material form was conducted might have affected the nuance of the comparison between the LMSs.

Another requirement where ambiguity has been detected is the requirement of a course overview from a course participant perspective. Here, the ability to see all courses in which the learner was enrolled was tested and evaluated. However, in relation to how the requirement was formulated, *An overview of all courses available to the course participant exists*, and in relation to the business requirements of sellability and marketing, this requirement might have been wrongly tested. Instead of testing whether the course participant could see all courses in which they were already enrolled, it might have been more valuable to test whether the course participant could see all courses *available* to them, as stated in the requirement. Since the course overview functionality in the tested LMSs varied, where in some LMSs only courses to which the learner had been invited to was visible, and in other LMSs all courses created by the client were visible, a more nuanced picture might have appeared regarding the course overview if the requirement would have been tested differently.

5.1.3 The requirements' alignment with theories on learning

Some of the requirements stated in this study might have an impact on the learning for future course participants, which calls for an analysis of whether these requirements have support in research on digital learning and learning in LMSs. The analysis and discussion of learning in relation to the requirements will contribute towards an answer on the research question *In which ways do the requirements on an LMS benefit the course participants' learning?*.

Three of the business requirements state that the LMS should be digital accessible, which is also highlighted by Sims (2021) who says that the accessibility needs must be taken into consideration. Furthermore, Sims (2021) pinpoints a critical aspect of testing the user-friendliness of an LMS, which is also reinforced by Sabharwal et al. (2018). This parallels well with the requirements of a user-friendly LMS without any severe issues. Another requirement relating to research in the field is the call for respons-

ive design to enable course participants to complete the courses on mobile platforms. This flexible functionality reduces the distance to learning according to Sabharwal et al. (2018). Further, the list of requirements does not include a requirement for the system language to correspond to the language used in the courses. This affects the user-friendliness when using the LMS (Little, 2015), and would thus have been a relevant requirement to test when evaluating the LMSs. The fact that using a separate system language might affect the experience was confirmed by one of the user experience testing participants who mentioned that it was annoying when the system language was in English and the course language was Swedish.

There are multiple sorts of functionality in the list of requirements that together creates flexibility when designing courses in an LMS. The need of having both modules and sections in a course is an example of a potentially effective way to reduce the complexity of a course by dividing the course material into smaller sections. This sectioning of material could contribute to avoiding the scenario of overflowing the cognitive capacity of learners (Clark & Mayer, 2016). Furthermore, Clark and Mayer (2016) recommends the use of varied content forms to engage the learner and then enable effective learning, which correspond well to the requirements of the possibility to use several content types. Since learner-content interaction is important when educating adults (Moore, 1989), it is important to evaluate the flexibility in creating course material. The flexibility could then enhance the possibility for learners to interact with the material in order to develop a deeper understanding of the material.

One requirement that was extracted from the interview data was the need for a customer admin to follow the progress of employees. The possibility to extract data of learners' progress is important in order to evaluate and develop courses (Bailey et al., 2018; Veluvali & Surisetti, 2022). However, this specific requirement is isolated to the role of a customer admin, and none of the employees of the client mentioned this when reflecting over requirements from a course creator point of view. The ability to trace customers was mentioned in the business requirements, but only to trace purchases of a course to improve sellability. Therefore, the list of requirements lacks a requirement concerning the possibility of collecting data, in order to help the course creators to secure high-quality courses in the LMS.

Several requirements are based on the needs of course participants, for example the possibility to repeat parts of or an entire course, which is important to enhance the

learners' ability to learn, as endorsed by J. R. Anderson (2010). Another course participant requirement was a discussion forum, where the learner could ask and answer questions. According to Veluvali and Surisetti (2022), a discussion forum can be a useful tool when collecting data in order to analyse the learning process of learners. However, in the interview data it was not the need for collecting data that drove the requirement of a discussion forum forward. Instead it was from the perspective of the course participant this requirement arose, to give the learner a possibility to discuss the material with others. This purpose of a discussion forum corresponds well to the interaction type learner-learner, which is according to T. Anderson (2003) not critical to have present, if one of the other interaction types is of high quality. In an LMS it might be easier to secure the interaction of learner-content (Moore, 1989). However, the requirement of a discussion forum is of low-priority, which corresponds well with the words of T. Anderson (2003). To enhance the learner-instructor interaction regular reminders can be used (Martin & Bolliger, 2018), which is also covered by the low-priority requirement of reminders in an LMS.

Another low-priority requirement was the functionality of generating digital badges for completed courses. The possibility of receiving a badge has been shown to improve the motivation for learning, however no correlation has been found between the use of badges and improved learning (Iwata et al., 2019). With that being said, one reason to use certificates that came up during the interviews was the possibility to share the certificate on social media and is therefore not a requirement that highlights improvement of the learning process. Instead of using badges to improve learning, reminders can be used, since weekly reminders can improve learners' results (Fulton et al., 2013). The functionality of weekly reminders is covered with the requirement of adding deadlines and/or reminders to a course.

According to previous research on how to improve digital learning, some important aspects are not covered by the list of requirements established in this study. For instance, no requirements speak about personalised feedback, which has been discovered to be a powerful tool to excel learning (Hattie & Timperley, 2007). The reason for this gap in the list might be because of the time resources that are needed to personalise feedback to course participants. Furthermore, the client has expressed that they want the participants to be able to complete courses without involvement from educators. However, it can be argued that the use of quizzes and progress bars can help the learner to receive answers on the three questions (a) "Where am I going?", (b) "How am I going?"

and (c) “Where to next?”, which according to Hattie and Timperley (2007) are important questions to be answered to satisfy the user’s need for feedback. To visualise the learner’s progress in a course is vital in order to make sure that the learner is aware of how much of the material they have yet to learn (Klerkx et al., 2014), and a progress bar can be a potential tool to implement this.

Another important aspect to consider when developing online courses is that it is important for the learner to access a course’s detailed curriculum and learning objectives (Bailey et al., 2018; Sims, 2021). The only requirement that speaks about the way a learner can view a course, is in the context of viewing all courses available for the learner. No requirements refers to the way the content for a specific course is visualised, nor setting learning objectives for a course. Additionally, another deficiency with the list of requirements is that it lacks requirements concerning functionality of the LMS with purpose to improve learners’ motivation to learn. An interesting possibility with an LMS is for example the use of gamification, which has been found to be an effective way to improve motivation (Hamari et al., 2014; Rawashdeh et al., 2021; Veluvali & Surisetti, 2022).

In summary, the comparison between the list of requirements and previous research shows that there are some critical aspects of digital learning, which are not included in the list. One of the most important aspects found is that the client do not mention the need of extracting statistics of course participants’ learning progress in order to develop their courses. To extract and analyse statistics to achieve high-quality courses is important according to several sources (Bailey et al., 2018; Veluvali & Surisetti, 2022), and therefore it should be considered when choosing an LMS. Furthermore, it is important for the course participants to receive personalised feedback (Hattie & Timperley, 2007), something that is also missing from the list of requirements. Lastly, there has been no mention of learning objectives and how the course should be visualised for course participants, which is a need highlighted by several studies (Bailey et al., 2018; Sims, 2021). The reason for these gaps in the list of requirements might be because of the economical resources the client has, or a lack of the interviewees’ experience. However, it can also be a consequence of gaps in the data set or missing questions in the interview guide. If questions would have been asked about these specific requirements the client might have agreed that these requirements are important to analyse in the testing of the LMSs.

5.2 Comparision between the LMSs

In order to answer the comparison aspect of the research question *How does a selection of LMS's meet the requirements in comparison to each other?*, it is important to understand how the different levels of priorities are valued. If an LMS scores low in the high priority requirements it might indicate that the LMS is not fit for the client. However, it might be relevant to evaluate if an LMS that scores high on all high priority requirements fails on most of the medium priority requirements. In a comparison with the other LMSs, this might indicate that the LMS is not a preferable choice for the client. Furthermore, the low priority requirements might not be relevant for the client to take into account when choosing an LMS, but for other companies that are interested in choosing an LMS, these requirements might be prioritised higher. Therefore, the results for the low priority requirements will still be discussed.

The high priority requirements are isolated to the business requirements, together with two requirements for the course creator role. AddWisdom was the only LMS that received the highest possible score on all high priority requirements. Thinkific and Teachable failed on the GDPR-requirement and Learnifier received low scores on multiple high priority requirements. If only taking the high priority requirement into account, AddWisdom would be a preferable choice when choosing an LMS.

However, AddWisdom scored very low in the evaluation of digital accessibility in comparison to the other LMSs, an area where Thinkific was the only LMS that scored high. In the context of this study, where a client with expertise in digital accessibility wants to find an LMS, this might be a breaking point. Nevertheless, accessibility is still a medium priority and for other companies this might not be a critical aspect when choosing an LMS. In the user experience tests from a course creator perspective, Teachable scored highest and Learnifier lowest. However, the scores were still similar and none of the LMSs had issues severe enough for the LMS to be disregarded. This aligns with the results of the user experience tests from a course participant perspective, where Thinkific scored the highest and Teachable the lowest, but without any severe issues in the LMSs. When it comes to medium prioritised requirements, Learnifier was the only LMS which had a material bank. However, none of the LMSs had well implemented synchronised updating, which was the reason to include the material bank requirement. The fact that Learnifier scored low on the requirement of including presentations in a course might be a problematic deficiency when developing courses in this LMS, espe-

cially in the transitioning phase from lectures to digital courses. Furthermore, when testing the requirements for a customer admin Teachable and AddWisdom scored high on both the requirements, whereas Learnifier had an inadequate overview of the course participants. In Thinkific no user role existed that could add new course participants to a course, except for the role of a course creator. This means that either Teachable or AddWisdom would be the preferable choice from a customer admin point of view.

As stated earlier, even though the low priority requirements are not prioritised for the client, they might be relevant for other companies. Learnifier was the only LMS where notification settings could be changed for a course creator, whereas a course participant could change the settings for notifications in all LMSs except for in AddWisdom. When testing if quizzes can be made mandatory some deficiencies were found in Teachable and Learnifier, which can be problematic if the client or other companies want to ensure that course participants have fulfilled the learning objectives for a course. In Learnifier and AddWisdom one could add deadlines and reminders. Further, in AddWisdom the requirement of a discussion forum was not met. In conclusion, Learnifier scored best when looking at only low priority requirements.

In summary, it can be distinguished that AddWisdom meets all high priority requirements and most other requirements on a satisfying level. On the other hand, AddWisdom did not score higher than the other LMSs on any medium priority requirements. Since most of the functionality requirements were found in the medium priority requirements, it is instead Thinkific or Teachable that would be preferable choices for a company prioritising functionality over business aspects. Learnifier unfortunately scored low on many of both the high and medium priority requirements in comparison to the others and might not be an option for any companies based on the list of requirements being transferable.

5.3 Method discussion

During the study, multiple methods were used for the different phases. In this section the used methods are evaluated in relation to the study and the achieved results. Circumstances and choices which might have affected the result for both data analysis and the comparative study are discussed and the effects on the results are estimated.

5.3.1 Data collection and analysis

In Chapter 3, a detailed description of how the semi-structured interviews were conducted is presented, which includes some possible pitfalls. As described there, the authors had the possibility to add questions to the interview guide between the interviews. This occurred between some of the interviews, when the authors wanted to deeper investigate an opinion expressed by one interviewee. The authors have concluded that this did not affect the results, since this was a qualitative study where the number of data points were not taken into account. Another method choice that could affect the result is that the interviews were not transcribed. However, during the analysis and when adding quotes to the results, the authors listened to the recorded interviews in order to avoid any mistakes in the interview notes.

The thematic analysis of the study was conducted in a both inductive and deductive manner. The reason for this combination was the authors inability to remain fully objective when coding the data set, because of the underlying purpose of finding requirements on an LMS and the authors existing thoughts and ideas about LMSs. This might have influenced the themes created from the codes, but on the other hand the interviews used to collect the data did also follow the same purpose. The purpose of finding requirements did also align with the purpose of the whole study, and therefore the entire study is influenced by that intention, which can be argued to be reasonable. However, one important aspect of the thematic analysis is that when the authors coded the data set, the codes from one interview might have affected the codes created from the next interviews, since one subconsciously tries to find patterns in the data. Therefore, the codes found immediately in the first interview analysed might occur more often. The authors tried to avoid this by starting reading different interviews and rereading the interview materials multiple times before coding the material.

5.3.2 Comparative study

The user experience testing from a course participant perspective was conducted by employees of the client's customers. When creating test groups the authors tried to create equivalent groups, but unfortunately this was not always possible. The group of people willing to participate in the testing was limited and the volunteers were not fully diverse in the different aspects taken into account when creating test groups. This might have affected the test results, since one test group had more UX-designers than

another test group, and the authors can not ensure that this did not affect the scores these LMSs received in the SUS evaluation. Furthermore, when performing a SUS evaluation it is important that the participants fill in the SUS form directly after testing the LMS (Brooke, 1995). It was not possible for the authors to ensure that the participants followed these instructions, which might have affected how the participants answered the questions, since they might have forgotten issues they experienced when testing the LMS. Additionally, in the LMSs it was possible for the authors to see that some of the participants tested the LMSs over a span of a few days instead of on a single occasion. Since a person might forget one thing they learned about the LMS one day, and had to learn it again the next day, this might have led to a lower score of the usability for that specific LMS. Furthermore, in the question form filled out by the participants they had the possibility to describe thoughts and opinions on issues they encountered during the testing. To describe an issue in text might be time consuming and tiresome for some people, which might have led them to skip this opportunity. To avoid missing important issues in the LMSs it might have been efficient to conduct follow up interviews with the participants or conduct video monitored testing sessions. On the other hand, this kind of interviews and/or sessions would have been very time consuming for both authors and participants and might have led to smaller test groups, causing a lower validity of the test results.

It was decided that the only users that would participate in the testing were the course participants and the client's CEO. The reason for this was that these two roles were deemed as the most important ones to collect input from. One disadvantage of this might however be that the views of the marketer, educator and UX-designer roles were reduced in the test result, since testing from these perspectives was only done by the authors. One of the main reasons not to include these roles in the testing phase was that there was a time-limit. Another reason was that the test cases which verified the wanted functionality were designed in a way which allowed yes/no answers. This should mean that who performs the test is of no importance for finding out whether the functionality exists or not. But since the test cases were evaluated with a number 0–3, along with a comment in the test protocol, some risk for subjective assessment existed even in the functionality testing. If the result of the tests cannot be assumed to be valid outside the study, the external validity of the study is lowered (Patino & Ferreira, 2018). Not letting the marketers, educators and UX-designers participate in the testing might thus have resulted in lowering the external validity of the study. Along with this, there might

be a risk that the authors were unable to find a wanted functionality in the LMS, even though it did exist. In that case, the LMS would not be fairly evaluated. However, if the functionality could not be found, that might indicate that it was poorly implemented in the LMS. The fact that the test protocol was carried out by both authors could lead to a reduced risk of missing functionality.

When testing the LMSs, the authors might have been influenced by experiences from testing the previous LMS, resulting in an unintended comparison already during the testing stage of the study. To avoid comparing the LMSs during testing, the authors went through all the requirements after the testing was concluded in order to identify scores that might be unreliable because of any unintended comparison. Furthermore, both authors have previous experiences and knowledge about software and LMSs which might have affected the evaluation of the LMSs. If the comparative study was repeated by others, the results might differ from the results of this study. One solution to get a less biassed result could have been to let employees at the client perform parts of the testing. However this was not possible because of the client's limited resources and time, and might therefore be advantageous for the client to perform supplementary studies before choosing an LMS.

One of the main reasons that the usability was only evaluated by real users from a course participant perspective is that this was considered as the most valuable perspective. Even though the in-house educators, marketers and UX-designers of the client might have valuable input on the usability of the product, they will eventually learn how to use the product since they will interact with it on a regular basis. The course participants might on the other hand only use the LMS once or very seldom, wherefore the usability aspects of the course participants become more important. If the course participant finds the system difficult to use, it might result in an unfinished course for that course participant, and in the end less revenue for the client.

Another critical aspect which might have affected the results of this study severely was the choice of excluding LMSs that did not offer free trial periods. In light of the limited resources of this study it was not possible to evaluate all LMSs included in the first selection. Another aspect of using free trial periods was that a lot of functionality was not included in the free packages and thus it was not always possible to distinguish which functionalities the LMS did in fact offer. With more funding and a longer time period it would have been possible to test more LMSs and more requirements, which

might have resulted in a more nuanced and generalisable result.

5.4 Conclusion

In this thesis, a list of requirements based on a small consulting firm's needs have been presented. Even though some method choices and circumstances of the study, such as the authors' biassed ideas and some deficiencies in the usability testing, the results of the study are considered relevant in relation to the purpose of the thesis. Many requirements can be transferred to other companies of similar size and interest as the client. However, which specific requirements to include and the prioritisation level of each requirement might differ when transferring the list to other companies. Also, some requirements should have been clarified before testing, or defined in other ways.

In comparison with research done on the area of digital learning and learning in LMSs, the list of requirements corresponds well to multiple areas, for example the possibility to divide course material in smaller sections and to include several material formats. The list of requirements is however partly insufficient in relation to digital learning research, since requirements concerning feedback possibilities, system language, functions supporting material improvement, motivation, visualisation of course material and learning objectives are missing.

The LMSs included in the study answers to the list of requirements in various ways. When analysing the test results and comparing the four LMSs to each other, it is evident that AddWisdom was the best performing LMS regarding the high priority requirements. AddWisdom would thus be the recommended LMS for the client to use. If looking only at the medium priority requirements, most of which relates to the LMS functionality, Thinkific or Teachable would be the preferable choices. In order to determine which of these two LMSs is the better choice, one has to prioritise between the different medium priority requirements where the LMSs scored differently. Learnifier did not prove to be a favourable choice in light of study's requirements and testing.

5.5 Directions for future research

Some aspects of this study would be interesting to analyse deeper to get a more nuanced and transferable answer to the research questions.

In order to recommend an LMS suitable for several different small consulting firms, it would be preferable to gather requirements for an LMS from a larger number of companies. This would most likely create a more transferable and comprehensive list of requirements that could be used by several companies in their pursuit to find a suitable LMS. It would also be preferable to expand the data set by conducting data collection at small consulting firms that are already teaching courses through an LMS. Moreover, the size of the test groups used in this study has been limited, and in order to get a more reliable SUS score, testing on a larger group of participants could be performed. Additionally, the number of LMSs tested in this study was limited, and it might be useful to evaluate additional relevant LMSs for a small consulting firm available on the market.

Furthermore, when the authors analysed previous research on digital learning and learning in LMSs, a gap was found in research on the area of short courses isolated to an LMS and the effects on adults learning when using LMSs at workplaces. This reflection is shared by Sabharwal et al. (2018) who conducted a larger literature study on the use of LMSs in workplaces and also found a deficiency in the research area. The authors of this thesis therefore recommend that future research should be conducted on the area of LMS usage in workplaces.

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Appendix A

Interview guide

Table A.1

Interview guide - general questions

Question
What is your role at the company?
What is your role in the education process at the company?
Have you completed any course at the company?
Do you have any prior experience with an LMS?
If so, what LMS? In what role? How was your experience?
What are your requirements and expectations on an LMS?
What potential advantages do you see when moving parts of a course to an LMS?
What potential disadvantages do you see when moving parts of a course to an LMS?
What expections and requirements do you think course participants will have?
If you would develop a course from scratch in an LMS, how would you design it? What are the most important aspects?
Do you have something more to add, something you think we have forgotten to ask about?

Table A.2

Interview guide - educator questions

Question
Is there a given strategy for how teaching should be performed?
What tools are used? For example powerpoint or trying-out exercises. What advantages or disadvantages have you noticed about using these tools?
What does you teaching look like?
Do you use any knowledge measurement in your teaching? For example tests or quizzes.
Are there any challenges in your teaching?
What kind of material do you think needs to be used in a digital course?
What would the purpose of using an LMS be for you?

Table A.3

Interview guide - education creator questions

Question
Is there a given strategy for how courses should be developed?
What kind of knowledge is supposed to be conveyed through your courses?
What are the learning objectives in the introductory course?
What kind of material do you think you should be able to use in a digital course?

Table A.4

Interview guide - CEO questions

Question
What does your education look like today?
Why do you want to start using an LMS?
What is the purpose of using an LMS?
Who are the potential participants of your digital courses?

Continues →

Question
What are the most important requirements on an LMS, according to you as CEO?

Appendix B

System usability scale

Table B.1

System usability scale - questions (Brooke, 1995)

Number	Question
1	I think that I would like to use this system frequently
2	I found the system unnecessarily complex
3	I thought the system was easy to use
4	I think that I would need the support of a technical person to be able to use this system
5	I found the various functions in this system were well integrated
6	I thought there was too much inconsistency in this system
7	I would imagine that most people would learn to use this system very quickly
8	I found the system very cumbersome to use
9	I felt very confident using the system
10	I needed to learn a lot of things before I could get going with this system

Appendix C

Test protocol

Table C.1

Course creator tasks in test protocol

Task
Create course creator profile
Create course *Course 1*
Create module *Module 1*
Add a section to *Module 1*
Add image, video, text, presentation and quiz to the section
Copy *Module 1* to a new course *Course 2*
Make changes to image and text in *Module 1* in *Course 2* and see if it is updated in *Course 1*
Add image to material bank and add it to *Module 1* in *Course 1*
Make changes to the image in the material bank
Add a certificate and/or badge to *Course 1*
Change the requirements on the certificate and/or badge
Make quizzes mandatory in *Course 1*
Turn on dark mode
Change the notification settings

Continues →

Task
Download *Course 1*
Add two more modules to *Course 2*
Add a discussion forum to *Course 2*
Add deadlines and/or reminders to *Course 2*
Publish the courses and share invitation to course participants
Make one of the course participants admin
Fill out the SUS-form

Table C.2

Admin tasks in test protocol

Task
Log in as admin
Add course participants to course
View course participants progress

Table C.3

Course participant tasks in test protocol

Task
Use the invite to create an user account
Look at the deadlines for *Course 2*
Start the course and complete one module
Log in and log out of the LMS
Look at the progress for *Course 2*
Complete *Course 2*
Go back to the first module in *Course 2*
Change the speed on the video in *Module 1*

Continues →

Task
Ask a question in the discussion forum
Download your certificate and/or badge for *Course 2*
Change the notification settings
Switch to dark mode
Go to *Course 2* and try to skip one module
Log in to your account on your mobile phone and complete *Course 1*

Appendix D

User instructions

Table D.1

Course participant tasks for usability testing

Task
Use the invite to *Introductury course in digital accessibility* to create an user account
Go to the course and complete some of the modules
Log in and log out of the LMS
Go back to the course and complete all the modules
When you have completed the course you are welcome to click around on the website
Fill out the SUS-form linked in the email

Appendix E

Test results

Table E.1

Business requirements with high priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
The client's costs of using the platform is a fraction of the revenue	1	3	3	3
Any employee of the client can handle the platform maintenance	3	3	3	3
The servers handling customer data are placed within the EU (GDPR requirement)	3	1	1	3
Individual user accounts can be created	3	3	3	3
A payment solution is integrated in the LMS, allowing customers to pay for courses without involvement from the client	1	3	3	3
Customers can sign up independently without help from the client	1	3	3	3
Customers can complete the course independently without help from the client	3	3	3	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, "-" - not possible to test.

Table E.2

Business requirements with medium priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
Courses are searchable for customers outside the client's customer base	0	0	0	0
Able to navigate with the help of keyboardScores high on a digital accessibility review	2	3	3	1
Good colour contrast	2	1	3	3
Possible to use screen narrator	1	2	2	1

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, "-" - not possible to test.

Table E.3

Business requirements with low priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
The design is adaptable to the client's theme used elsewhere	3	3	3	3
The client is able to see all customers and trace course purchases	3	3	3	3
The LMS is integrable with other commonly used LMSs	-	-	-	-

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, "-" - not possible to test.

Table E.4

Course creator requirements with high priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
Courses can be divided into modules	3	3	3	3
Modules can be divided into sections	3	3	3	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.

Table E.5

Course creator requirements with medium priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
Created modules can be used in multiple courses and be updated simultaneously	0	0	0	0
A common material bank exists, where updates on the material is shown in all courses using the material	2	0	0	0
Visual interface rather than text- or code-based	3	3	3	3
Images	3	3	3	3
Videos	3	3	3	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
Text	3	3	3	3
Presentations	1	3	3	3
Quizzes	3	3	3	3
Scores high on System Usability Scale	76	96	82	87

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.

Table E.6

Course creator requirements with low priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
Notifications settings can be changed	3	0	0	0
Dark mode can be turned on	0	0	0	0
An entire course or parts of it can be downloaded	0	0	0	0
Certificates and/or badges can be added to a course	3	3	3	3
Requirements for acquiring a certificates and/or badges are changeable	2	2	2	2
Quizzes can be made mandatory	2	2	3	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
Deadlines can be added to a course	3	0	0	2
Auto-generated reminders can be sent out to course participants	3	0	0	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.

Table E.7

Course participant requirements with medium priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
Scores high on System Usability Scale	83	73	91	80
The course participant’s process in a course is visualised	3	3	3	3
The course participant’s progress is saved	3	3	3	3
The course participant can go back to earlier modules	3	3	3	3
The course participant can repeat completed courses	3	3	3	3
An overview of all courses available to the course participant exists	3	3	3	3
The LMS is compatible with mobile platforms	3	3	3	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.

Table E.8

Course participant requirements with low priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
The playback speed of videos can be changed	3	3	3	3
Notification settings can be changed	3	3	3	0
Dark mode can be chosen	0	0	0	0
Questions can be asked and discussions can be started in a forum	3	3	3	1
A certificate and/or a badge can be generated for a course	3	-	-	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.

Table E.9

Customer admin requirements with medium priority

Requirements	Learnifier	Teachable	Thinkific	Addwisdom
A detailed overview of course participants and their progress in different courses exists	1	3	3	3
New employees can be added and given access to different courses	3	3	0	3

0 - could not be found, 1 - barely usable, 2 - has some flaws, 3 - implemented in a satisfying way, ”-” - not possible to test.